

PORTFOLIO PROJECT

Employee Attrition & Retention Analytics

End-to-end people analytics and business intelligence

A reproducible portfolio solution that transforms a fictional HR practice dataset into governed Delta tables, validated analytical outputs and a four-page Power BI report.

Databricks

PySpark

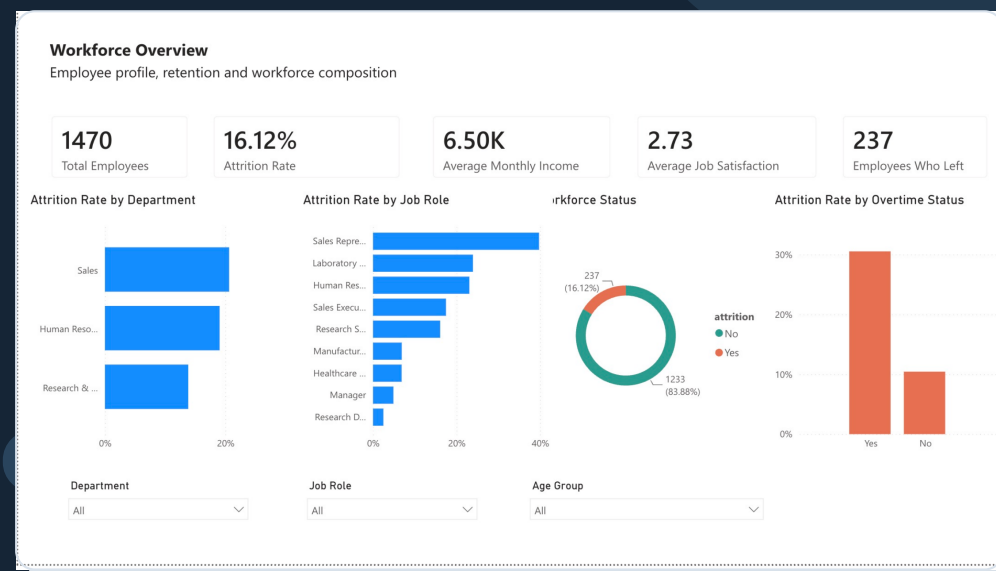
Spark SQL

Delta Lake

Power BI

Valtteri Paivinen

Data and BI portfolio | 2026



Project at a glance

Business context, scope and measured outcomes



Business objective

- Measure workforce attrition and retention at an aggregate level.
- Identify departments, job roles and employee groups for deeper HR investigation.
- Explore overtime, travel, satisfaction, income, tenure and career-development patterns.
- Deliver an interactive, documented and ethically framed Power BI portfolio solution.

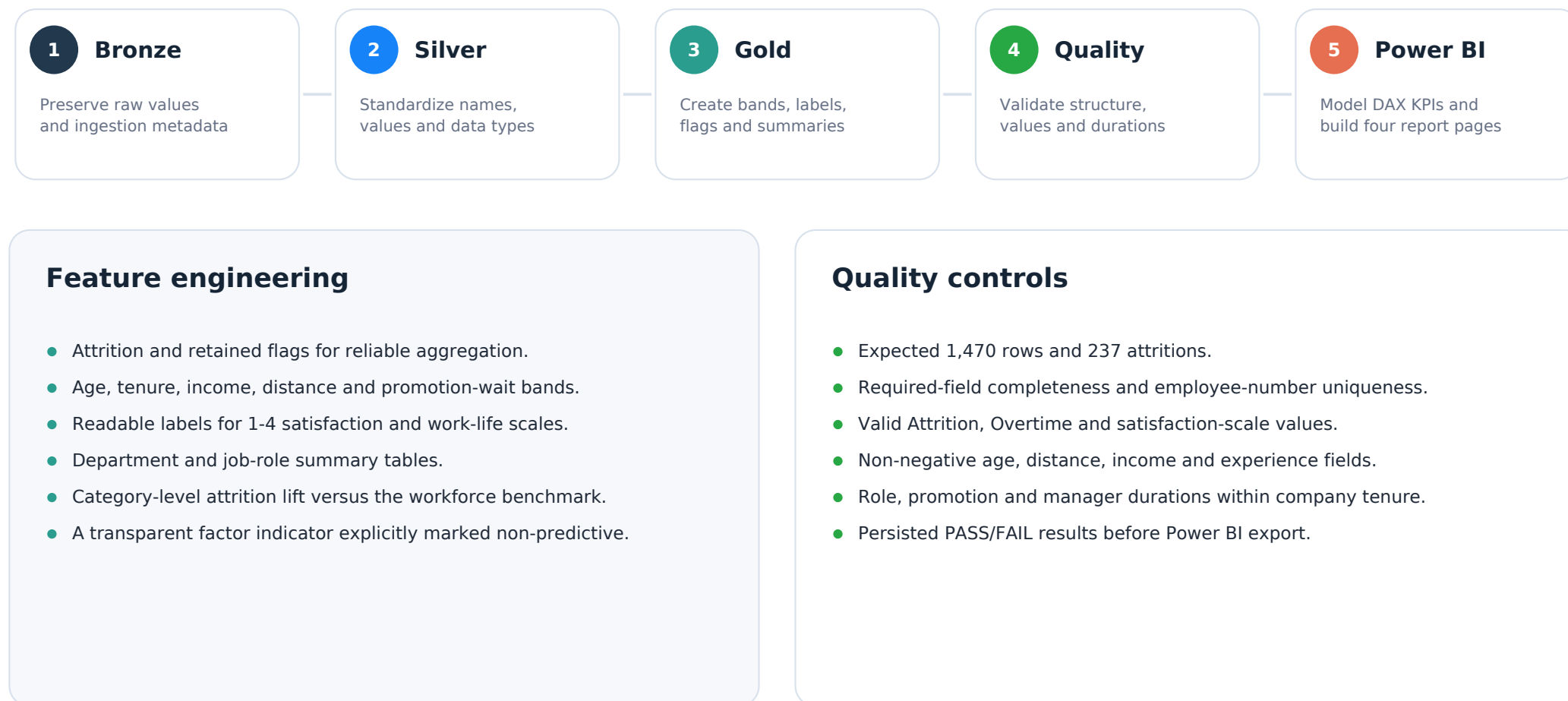
What the solution delivers

Layer	Deliverable
Lakehouse	Bronze, Silver, Gold and quality Delta tables
Analysis	Reusable SQL and category benchmarks
Semantic model	Reusable DAX measures for HR KPIs
Reporting	Four interactive Power BI pages
Documentation	README, data dictionary and visual overview

Source context: fictional IBM-created practice data; results do not describe IBM's real workforce.

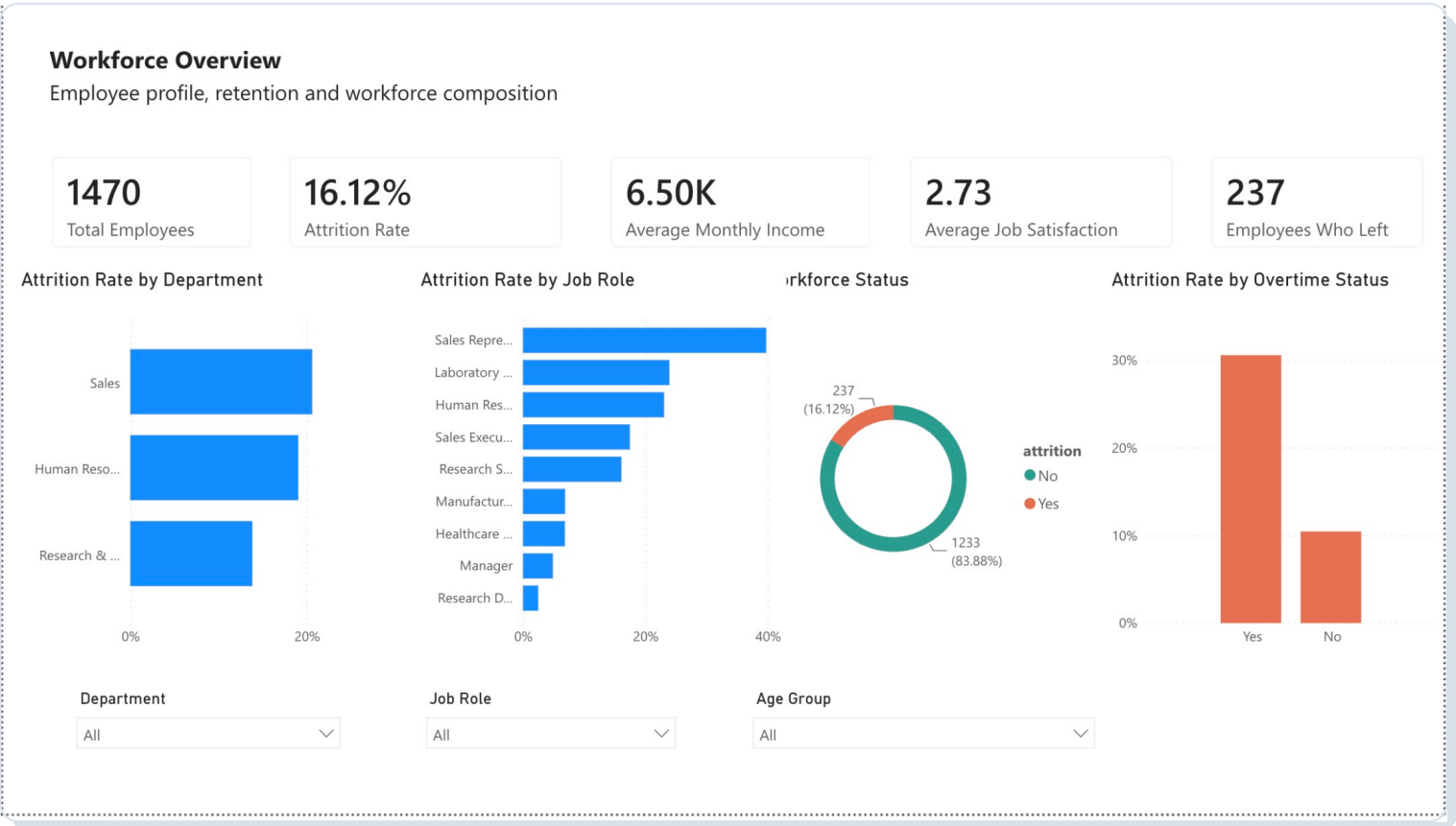
From source file to decision-ready reporting

A reproducible lakehouse pipeline with validation and semantic modelling



Workforce Overview

Employee profile, retention and workforce composition



Analytical focus

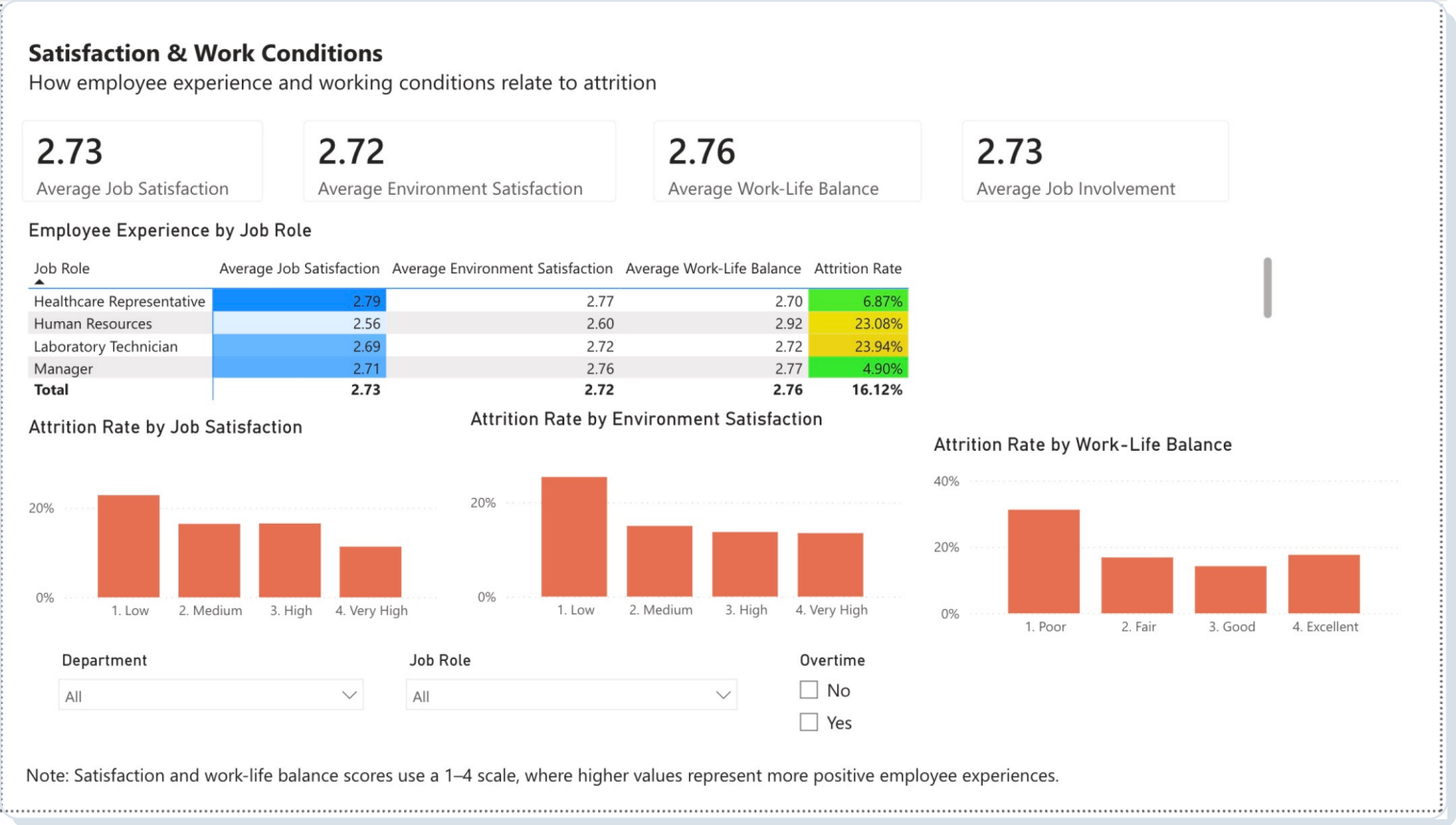
- Executive workforce KPIs
- Department and job-role comparison
- Retained-versus-left composition
- Overtime comparison and slicers

What this page reveals

- 1,470 employee records
- 237 employees left
- 16.12% overall attrition
- 30.53% overtime attrition

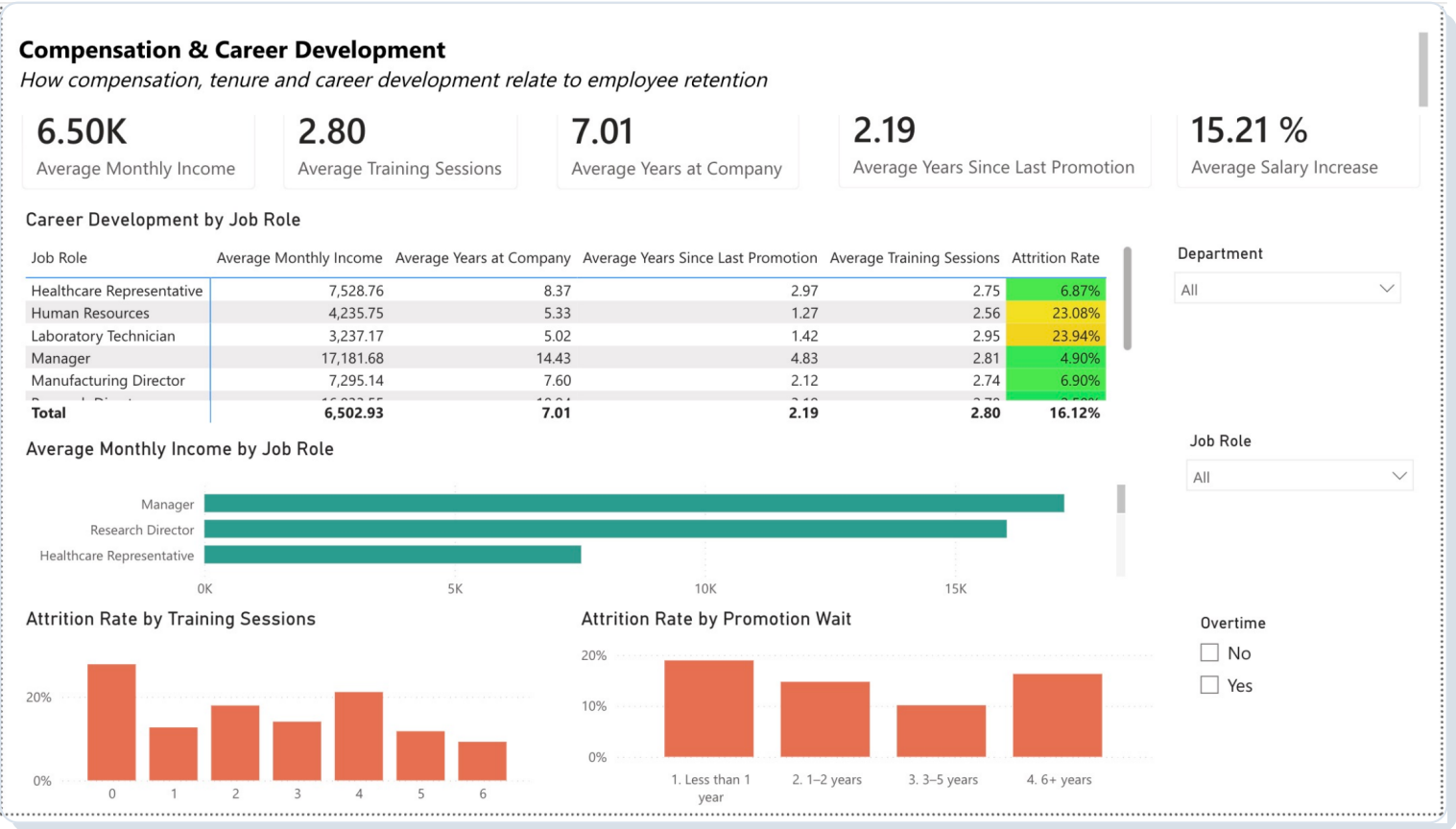
Satisfaction & Work Conditions

How employee experience and working conditions relate to attrition



Compensation & Career Development

How compensation, tenure and career development relate to retention



Analytical focus

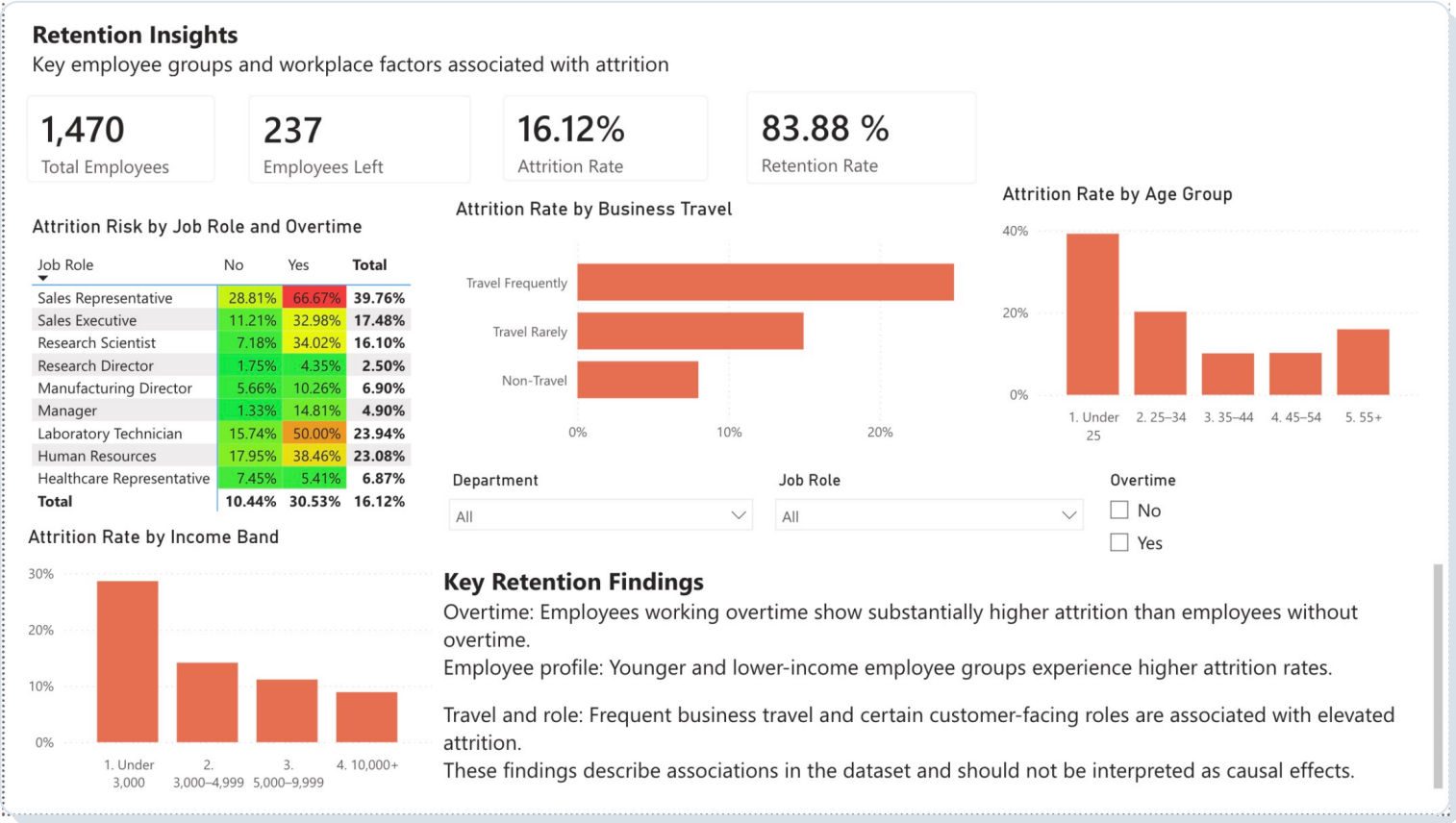
- Income and tenure KPIs
- Career metrics by job role
- Training-session comparison
- Promotion-wait comparison

What this page reveals

- Average income: 6,502.93 units
- Average tenure: 7.01 years
- Average promotion wait: 2.19 years
- Average salary increase: 15.21%

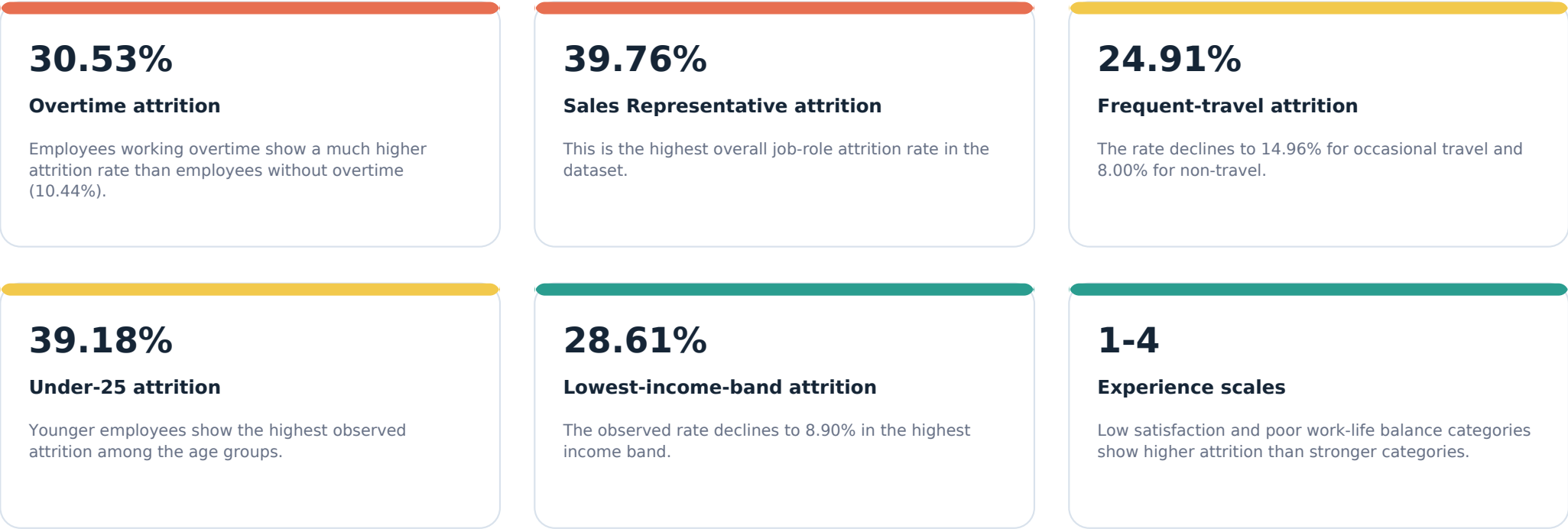
Retention Insights

Key employee groups and workplace factors associated with attrition



Decision-support findings

Observed patterns to investigate — not causal conclusions



Recommended HR follow-up

- 1

Review overtime workload, staffing levels and manager practices in the highest-attrition roles.
- 2

Investigate Sales Representative retention with interviews and role-specific workload data.
- 3

Examine travel frequency, recovery time and flexibility for regularly travelling employees.
- 4

Strengthen onboarding and career support for younger and lower-income employee groups.

A portfolio project built for transparent analysis

Reproducible workflow, clear limitations and responsible HR framing

How to reproduce

- 1 Download the fictional HR practice dataset described in `data/README.md`.
- 2 Upload the CSV to the Databricks source volume.
- 3 Run notebooks 01-04 in numerical order.
- 4 Confirm MODEL STATUS: PASS in the quality table.
- 5 Import the four curated CSV exports into Power BI.
- 6 Create the documented DAX measures and four report pages.

Responsible-use boundaries

- The dataset is fictional and does not describe IBM's real workforce.
- The source has no date field, so no trends or forecasts are fabricated.
- Income values have no documented currency and are shown as units.
- Observed associations do not establish causal effects.
- The factor indicator is not a predictive model and cannot be used to rank employees.
- Production use would require privacy review, fairness monitoring and human oversight.

Project deliverables

Four Databricks notebooks • reusable Spark SQL • documented DAX measures • four Power BI dashboard pages • data dictionary • GitHub README • visual project overview

Valtteri Paivinen github.com/valtteri-paivinen | linkedin.com/in/valtteri-paivinen